Giries Abu Ayoub - Curriculum Vitae

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Experience

Camtek August 2026 - Present

Applied AI Researcher *Migdal Haemek, Israel*

- Worked on open computer vision and deep learning problems for semiconductor inspection and metrology.

University Of Haifa May 2024 - July 2026

Assistant Researcher - CV lab *Haifa, Israel*

- Investigated representation learning and multimodal alignment (Visual-Text-Audio) in zero-shot/low-shot settings. Developed the SpurAudio benchmark to evaluate spurious correlations. Optimized large backbones such as Dinov2 and CLIP for fine-grained classification.

University Of Haifa March 2023 - July 2026

Teaching Assistant *Haifa, Israel*

- Lead tutor on Linear Algebra course (7 semesters)
- Lead tutor on Deep Learning / Computer Vision course (3 semesters)
- Lead tutor on Intro to CS in python course (2 semesters)
- Tutor for Intro to C course (2 semesters)

Auura inc. June 2025 - October 2025 (Internship)

Machine Learning / Backend Engineer - python, docker *Haifa, Israel*

- Developed an AI enhanced web crawler to exctract funding opportunities, including link discovery, document parsing, content classification and recognition tasks.

University Of Haifa September 2022 - August 2023

Software Developer *Haifa, Israel*

- I worked on numerous projects while at the university's Robotics Lab using SLAM and navigation algorithms, such as PnP, RRT, also contributed to the lab's simulator to simulate the Algorithms virtually on linux and windows systems

Education

University Of Haifa March 2023 - July 2026

M.Sc in Computer Science (GPA: 94) *Haifa, Israel*

- Thesis: Few-Shot Feature Adaptation via Balanced Pairwise Affinities on Frozen Vision Transformers - Link
  Advisor: Dr. Simon Korman

University Of Haifa April 2019 - February 2023

B.Sc in Computer Science (GPA: 84) *Haifa, Israel*

- Object Oriented Programming(96), Data Structures(83), Operating Systems(86),Software Engineering(93), Computer Vision(93), Computer Graphics(100), Linear Algebra (100), Probabilistic Algorithms (86)

Publications

- Supportive Token Revealing for Fast Diffusion Language Model Decoding
  AACL-IJCNLP 2026 ORAL - Link
- SpurAudio: A Benchmark for Studying Shortcut Learning in Few-Shot Audio Classification
  Under review Link
- SPECTRA: Subspace-Preserving Embedding Calibration, Transport, and Replay for Fully Few-Shot Class-Incremental Audio Classification
  Under review Link
- `SumReason`: A Benchmark for Reasoning-Driven Query-Conditioned and Standard Video Summarization
  Under review

Academic Projects

Strand Generation | *Pytorch, Generative AI*

- Built a dataloader to research different possibilities of protein structures using ESM3; used by the bioinformatics lab at the University of Kansas Git: Repo

PianoLearner | *Unity,C#*

- An MR application in VR using metaquest 2 and Unity as part of the MixedReality Lab, teaches the player how to play piano virtually without the need for an actual piano. Git: Repo

Cinema App | *Java, Maven, SceneBuilder, Eclipse*

- Full Stack Development of a Cinema System with full functionality and statistics. Git: Repo

Efficient ML: LLM and Diffusion Data Curation | *Pytorch*

- Explored reward-free subset selection strategies for fine-tuning diffusion generative models
- Built efficient subset selection framework for LLM finetuning

Hand Held NeRF | *Pytorch*

- Model a hand held object using Neural Radiance Fields (NeRF), and fixing circular GIFs using NeRF Git: Repo

Technical Skills

Languages: C, C++, C#, Python, Bash
Technologies: Unix based DGX Deep Learning Clusters, Docker, Google Colab
Environments: Opencv, Opengl, Pytorch, Tensorflow, Linux

Achievements

- Award + grant award for ”Excellence in teaching assistance” (2022-2023) during the Akavia ceremony
- Grant award for ”Excellent M.Sc research students” (2023 - 2024)

Social Engagements

- Bi-Weekly Deep Learning Group Discussions - University Of Haifa
- Haifa C++ meetup - Matam
- Reviewer - ICASSP, ICML, Neurips, AAAI